$= (1-\theta+1$

HW 8 Due 12/6

1) 1. $P(W<1) = 0.02$ light $n = 50$

$P(1 \le W \le 1.072) = ?$ good 0.9 X is # light boxes

$P(W > 1.072) = 0.08$ heavy Y is # good boxes

Trinomial Distribution: we have a sequence of n experiments with 3 possible mutually exclusive outcomes.

$$\frac{50!}{x!\,y!\,(50-x-y)!}(0.02)^x(0.9)^y(0.08)^{50-x-y}$$

$x, y \ge 0$

$x+y+z = 50$

$0 \le x+y \le n$

2. Binomial random variable, where $n = 50$

(Y or not) $p = 0.90$

3. $X = 3$ $Y|X=3$ $n = 47$

47 Y or Z $b(47, \frac{0.9}{0.98})$

↙ ↘

0.9 0.08

4. $E(Y|X=3) = 47 \cdot \frac{0.90}{0.98} = 43.163$

5. $\frac{Cov(X,Y)}{\sqrt{Var(X)\cdot Var(Y)}}$

5. $\sum_{x=0}^{50}\sum_{y=0}^{50-x} \frac{50!}{x!\,y!\,50-x-y!}(0.02)^x(0.9)^y(0.08)^{50-x-y}$

$\frac{0.018}{0.098}$

textbook formula; conditional means are $E[Y|x] = (n-x)\frac{p_y}{1-p_x}$

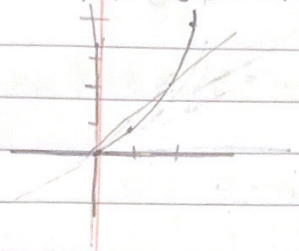
Since conditional means are linear, $E[X|y] = (n-y)\frac{p_x}{1-p_y}$

$$\rho^2 = \left(\frac{-p_y}{1-p_x}\right)\left(\frac{-p_x}{1-p_y}\right) = \frac{p_x p_y}{(1-p_x)(1-p_y)} \qquad \rho = -\sqrt{\frac{p_x p_y}{(1-p_x)(1-p_y)}} = -\sqrt{\frac{0.02\cdot 0.9}{(0.98)(0.1)}}$$

x and y are negatively correlated, $n = 50$, so as x $= -0.429$

increases, y decreases and vice versa.

2) 1. $\iint cxy\,dx\,dy = 1$ $\quad \int_{x=0}^{1}\int_{y=x^2}^{x} cxy\,dy\,dx = 1$



$$\int_{x=0}^{1}\left[\frac{cxy^2}{2}\right]_{y=x^2}^{x} dx$$

$$\int_{x=0}^{1}\frac{cx^3}{2}-\frac{cx^5}{2}\,dx = c\int_{x=0}^{1}\frac{x^3-x^5}{2}\,dx =$$

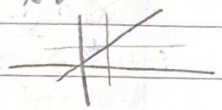
$$\frac{c}{24}=1$$

$$c=24$$

$$= \frac{c}{2}\left(\int_{x=0}^{1}(x^3-x^5)\,dx\right) =$$

$$= \frac{c}{2}\left[\frac{x^4}{4}-\frac{x^6}{6}\right]_{x=0}^{1} = \frac{c}{2}\left(\frac{1}{4}-\frac{1}{6}\right) = \frac{c}{24}$$

2. $f(x,y) = c(1+x^2y) \quad 0\le x\le y\le 1$



$$\int_0^1\int_0^{x=y} c(1+x^2y)\,dx\,dy = c\int_0^1\left[x+\frac{x^3y}{3}\right]_0^{x=y} dy = c\int_0^1\left(y+\frac{y^4}{3}\right)dy =$$

$$c\cdot\frac{17}{30}=1$$

$$c=\frac{30}{17}$$

$$c\left[\frac{y^2}{2}+\frac{y^5}{15}\right]_{y=0}^{1} =$$

$$c\left(\frac{1}{2}+\frac{1}{15}\right) = c\left(\frac{17}{30}\right)$$

3. $\int_0^1\int_0^{x=y^2} cye^x\,dx\,dy = \int_0^1\left[cye^x\right]_0^{x=y^2} dy = c\int_0^1(ye^{y^2}-y)\,dy = c\left(\int_0^1 ye^{y^2}\,dy - \int_0^1 y\,dy\right) =$

c

$u=y^2 \quad \frac{du}{dy}=2y \quad c\left(\int_0^1\frac{e^u}{2}\,du - \left[\frac{y^2}{2}\right]_0^1\right) = c\left(\left[\frac{e^u}{2}\right]_0^1 - \left[\frac{y^2}{2}\right]_0^1\right) =$

$$c\left[\left(\frac{e}{2}-\frac{1}{2}\right)-\left(\frac{1}{2}-0\right)\right] =$$

$$\frac{du}{2y}=dy$$

$$\frac{ce-2c}{2}=1$$

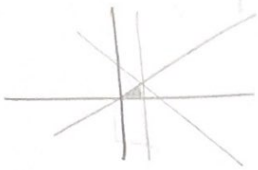
$$ce-2c=2$$

$$c(e-2)=2$$

$$c=\frac{2}{e-2}$$

$$c\left(\frac{e-1}{2}-\frac{1}{2}\right) = c\left(\frac{e-2}{2}\right)$$

$\int_0^{\frac{1}{4}} \int_0^{x}$



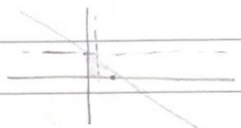


3) 1. $\int_0^1 \int_0^{x=1-y} cx(1-y)\,dx\,dy = c\int_0^1 \left[\frac{x^2}{2}\right]_0^{1-y}(1-y)\,dy = \int_0^1 \frac{(1-y)^2}{2}(1-y)\,dy$

$\frac{c}{2}\int_0^1 (1-y)^3\,dy = \frac{c}{2}\int_0^1 (1-2y+y^2)(1-y)\,dy = \frac{c}{2}\int_0^1 1-2y+y^2-y+2y^2-y^3\,dy$

$= \frac{c}{2}\int_0^1 1-3y+3y^2-y^3\,dy = \frac{c}{2}\left[y-\frac{3}{2}y^2+y^3-\frac{y^4}{4}\right]_0^1 =$

$\frac{c}{2}\left(1-\frac{3}{2}+1-\frac{1}{4}\right) = \frac{c}{8}$ $\quad \frac{c}{8}=1$ $\quad c=8$



2) $P(Y<X \mid X\le \frac{1}{4}) = \dfrac{P(Y<X, X\le \frac{1}{4})}{P(X\le \frac{1}{4})}$

$P(X\le\frac{1}{4})$ $\int_0^{1/4}\int_0^{1-x} 8x(1-y)\,dy\,dx = \int_0^{1/4}\int_0^{1-x} 8x(1-y)\,dy\,dx = \int_0^{1/4} 8x\left[y-\frac{y^2}{2}\right]_0^{1-x}dx = \int_0^{1/4} 8x\left(1-x-\frac{(1-x)^2}{2}\right)dx =$

$= \int_0^{1/4} 8x\left(1-x-\frac{1}{2}+x-\frac{x^2}{2}\right)dx = \int_0^{1/4} 8x\left(\frac{1-x^2}{2}\right)dx = \int_0^{1/4} 4x(1-x^2)\,dx = 4\int_0^{1/4}(x-x^3)\,dx =$

$4\left[\frac{x^2}{2}-\frac{x^4}{4}\right]_0^{1/4} = 4\left(\frac{1}{32}-\frac{1}{1024}\right) = 4\left(\frac{32-1}{1024}\right) = \frac{31}{256}$

$P(Y<X, X\le\frac{1}{4})$

$\int_0^{1/4}\int_0^{x} 8x(1-y)\,dy\,dx = \int_0^{1/4} 8x\left[y-\frac{y^2}{2}\right]_0^{x}dx = \int_0^{1/4} 8x\left[x-\frac{x^2}{2}\right]dx$

$= \int_0^{1/4}(8x^2-4x^3)\,dx = \left[\frac{8}{3}x^3-x^4\right]_0^{1/4} = \frac{1}{24}-\frac{1}{256} \approx 0.0377$

$\frac{0.0377}{\frac{31}{256}} = 0.311$

4) 1. $f_x(x) = \int_0^{y=x} 2\,dy = 2y\Big]_0^x = 2x$ $\qquad f_x(x) = 2x,\ 0\le x\le 1$

$f_y(y) = \int_{x=y}^{x=1} 2\,dx = 2x\Big]_y^1 = 2-2y$ $\qquad f_y(y) = 2-2y,\ 0\le y\le 1$

$M_x = \int_0^1\int_0^x 2x\,dy\,dx = \int_0^1 2x^2\,dx = \left[\frac{2}{3}x^3\right]_0^1 = \frac{2}{3}$

$M_y = \int_0^1\int_y^1 2y\,dx\,dy = \int_0^1 2y-2y^2\,dy = \left[y^2-\frac{2}{3}y^3\right]_0^1 = \frac{1}{3}$

$\sigma_x^2 = E(X^2)-(E(X))^2 = \int_0^1\int_0^x 2x^2\,dy\,dx = \int_0^1 2x^3\,dx \hat{=} \left[\frac{x^4}{2}\right]_0^1 \hat{=} \frac{1}{2}-\frac{4}{9} = \frac{1}{18}$

$\frac{4}{9}$ $\qquad \frac{4}{9}$ $\qquad -\frac{4}{9}$

$\frac{1}{9}$ $\qquad$ $-\frac{1}{9}$

$\sigma_Y^2 = E(Y^2) - (E(Y))^2 = \int_0^1 \int_y^1 2y^2 \, dx\, dy = \int_0^1 2y^2 - 2y^3 \, dy = \left[\frac{2}{3}y^3 - \frac{y^4}{2}\right]_0^1 - \frac{1}{9} =$

$\frac{2}{3} - \frac{1}{2} - \frac{1}{9} = \frac{1}{6} - \frac{1}{9} = \frac{3}{54} = \boxed{\frac{1}{18}}$

$\frac{2}{3} \cdot \frac{1}{3}$

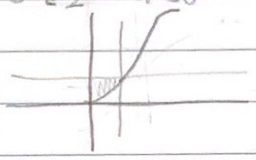
$COV(X,Y) = E(XY) - (E[X] \cdot E[Y]) = \int_0^1 \int_y^1 xy \cdot 2 \, dx\, dy - \frac{2}{9} = \int_0^1 2y \left[\frac{x^2}{2}\right]_y^1 dy - \frac{2}{9} =$

$\int_0^1 2y\left(\frac{1}{2} - \frac{y^2}{2}\right) dy - \frac{2}{9} = \int_0^1 (y - y^3)\, dy - \frac{2}{9} = \left[\frac{y^2}{2} - \frac{y^4}{4}\right]_0^1 - \frac{2}{9} = \frac{1}{2} - \frac{1}{4} - \frac{2}{9} = \frac{2}{8} - \frac{2}{9} =$

$\frac{18-16}{72} = \boxed{\frac{1}{36}}$

$\rho = \frac{COV(X,Y)}{\sqrt{\frac{1}{12}}\sqrt{\frac{1}{18}}} = \frac{\frac{1}{36}}{\frac{1}{18}} = \frac{18}{36} = \boxed{\frac{1}{2}}$

5) $\int_0^1 \int_{y=x^3}^{y=x} \frac{4}{3} \, dy\, dx = \frac{4}{3}\int_0^1 [y]_{x^3}^{x} \, dx = \frac{4}{3}\int_0^1 (x - x^3)\, dx = \frac{4}{3}\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 =$ $\quad x > y$



$\frac{4}{3}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{4}{3}\left(\frac{1}{4}\right) = \boxed{\frac{1}{3}} = P(X > Y)$

6) 1. $\frac{108-110}{10} = \boxed{-1/5}$ $\qquad$ $\frac{126-110}{10} = \boxed{8/5}$ $\qquad$ $P(-0.2 < Z_Y < 1.6)$

$P(Z_Y < 1.6) - P(Z_Y < -0.2) = 0.9452 - (1 - 0.5793) = \boxed{0.5245}$

2. $P(108 < Y < 126 \mid X = 3.2) =$

Mean: $110 + 0.6\left(\frac{10}{1.4}\right)(3.2 - 2.8) = 116$

SD: $10\sqrt{1 - 0.6^2} = 8$

$P\left(\frac{108-116}{8} < \frac{Y-116}{8} < \frac{126-116}{8} \,\middle|\, X = 3.2\right) = P\left(-1 < \frac{Y-116}{8} < \frac{5}{4} \,\middle|\, X = 3.2\right)$

$= P(Z_Y < 1.25) - P(Z_Y < -1) = 0.8944 - (1 - 0.8413) = \boxed{0.7357}$